

# PRAJWAL KUMAR K

Bengaluru, India | +91-9740167850 | [prajwal.k141205@gmail.com](mailto:prajwal.k141205@gmail.com)  
[LinkedIn](#) | [Portfolio](#) | [GitHub](#) | [Codeforces](#) | [CodeChef](#) | [LeetCode](#) | [AtCoder](#)

## EDUCATION

### Indian Institute of Information Technology, Kottayam

B.Tech in Computer Science and Engineering; CGPA: 8.66/10.0

Kottayam, Kerala

Aug 2024 – May 2028 (expected)

## TECHNICAL SKILLS

**Programming:** Python, C++, Rust, MATLAB

**AI/ML:** PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Matplotlib

**Research and engineering:** Jupyter, Simulink, FastAPI, Next.js, SQL

**Research interests:** Game theory, multi-agent learning, algorithms and systems, empirical AI evaluation

**Spoken Languages:** English, German, Hindi, Kannada

## EXPERIENCE AND LEADERSHIP

### Openverse and Coders Club

IIIT Kottayam

- Openverse Technical Lead (May 2025–Jul 2026): reviewed pull requests across the club and coordinated development.
- Coders Club Coordinator (Jan 2025–Present): curate problems, configure Codeforces Gym, and run campus contests.

## PROJECTS

### PokeForge | Python, PyTorch, Gen 9 OU

[Code](#)

- An agent-evaluation testbed for competitive Generation 9 OU Pokémon battles (in progress).
- Integrated pinned FoulPlay and Metamon baselines in isolated Pokémon Showdown environments; conformance and competitive strength remain under evaluation.

### Bluff | Imperfect-information games, opponent modeling

[Code](#)

- A repeated-play imperfect-information card game for studying bluff detection and adaptation to opponents.
- Built Bayesian opponent profiles and a learned public-information bluff detector; removing the detector lowered draw-aware score by 4.9–5.8 percentage points across eight synthetic profiles.

### Drishti-XAI | Retinal imaging, MATLAB, Python

SIH 2026 team submission

- An explainable retinal-screening prototype evaluated for generalization across diabetic-retinopathy image datasets.
- Built the MATLAB/Python prototype and evaluation pipeline; at a fixed model and threshold, sensitivity/specificity were 0.908/0.877 on APTOS (n=733) and 0.363/0.865 on Messidor-2 (AUC 0.666).

### KPlaceNet | Image geolocation, PyTorch, ResNet-50

[Code](#)

- A team computer-vision project that predicts geographic locations from images.
- Evaluated 12 ResNet-50 configurations on a spatially separated 3,000-image OSV-5M test subset; the best placed 3.33% within 200 km.

### BattleCP | Rust, Axum, Tokio, WebSockets

[Live](#) | [Code](#)

- A multiplayer Battleship game where solving Codeforces problems earns players turns.
- Built an Axum/Tokio WebSocket backend with server-authoritative game rules, submission validation, and reconnect support.

## ACHIEVEMENTS AND CERTIFICATIONS

**Publication:** First-listed author and presenter, “Collaboration-Enabled Energy-Aware Learning Model Exchange (CE-LME) Framework for Edge Computing-Assisted Applications,” IEEE ICICI 2025. Paper.

**Algorithmia:** selected for OOSC 4.0; team presentations at OOSC 3.0 (IIT Kanpur) and T4E 2025 (IIT Madras).

**Hackathon:** Best Software Award, IEEE MACE .hack(); 2026, Team Hogorithm (Anchor).

**Competitions:** Codeforces Pupil (peak 1221); CodeChef peak 1463; HTB HOLMES CTF: 86th/7,085, Team SuKMaDhe (65/65).

**Cyfrin Updraft (2025):** Blockchain Basics, Rust Programming Basics, Solidity Smart Contract Development, and Fundamentals of Zero-Knowledge Proofs.